

1 Perceived Stress Tracks the Tinnitus Symptom More
2 Than the Audiometric Threshold: A Survey-Weighted
3 KNHANES Study (STARS)

4 With a Prespecified Cross-National Validation Plan and a Clinician-Governed
5 Referral-Safety Component

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Note. The primary KNHANES perceived-stress analysis is reported as a complete empirical result; the cross-national validation, AI-extraction, and clinical-extension components remain prospective (design only).

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Abstract

Purpose: STARS tests, in nationally representative survey data, whether *perceived stress* is associated with the tinnitus *symptom* and with the audiometric *threshold*, and whether these associations dissociate. Analyses, variables, cycles, and adjustment sets were prespecified to limit analyst degrees of freedom. Because the surveys carry no validated occupational-stress instrument, the exposure is labeled perceived stress (a single general item), not occupational stress, and every estimate is an association, not a cause.

Method: KNHANES V (2010–2012; adults 40–69) is the primary cohort because it carries a general perceived-stress item; U.S. NHANES (2011–2018) provides supporting analyses under a PHQ-9 distress proxy. A directed acyclic graph fixes the minimal sufficient adjustment set (age, sex, occupational noise) and separates confounders

from mediators. Tinnitus and better-/worse-ear pure-tone average are modeled with survey-weighted (Taylor-linearized) regression, per STROBE.

Results (KNHANES): High perceived stress was associated with *tinnitus* (OR 1.42, 1.26–1.60) but only weakly and non-significantly with the *audiometric threshold* (OR 1.18, 0.99–1.41), which was dominated by noise and age—the prespecified dissociation of hypothesis H1. The tinnitus association survived adjustment for depressed mood and audiometric hearing loss and persisted for non-bothersome tinnitus. NHANES showed the same directional pattern (directional consistency, not validation). A cross-national validation plan and a clinician-governed AI/referral-safety component are prospective (no results).

Conclusions: In representative Korean adults, perceived stress tracks the tinnitus symptom but not the audiometric threshold once noise and age are accounted for—consistent with a partly central, non-cochlear correlate of tinnitus. Findings are reproducible from committed code; because the design is cross-sectional, directionality (including reverse causation) is unresolved, and prospective cohorts remain necessary.

Keywords: tinnitus; bothersome tinnitus; hearing loss; audiometry; perceived stress; work-related factors; KNHANES; NHANES; survey-weighted analysis; reproducibility; audiology.

1 Introduction

Tinnitus and hearing loss are among the most common auditory health concerns worldwide, affecting communication, sleep, emotional well-being, work participation, and quality of life. The World Health Organization estimates that more than 1.5 billion people live with some degree of hearing loss, a number projected to rise substantially by 2050 ([World Health Organization, 2021](#)). Tinnitus alone affects an estimated 10–15% of adults, and a clinically meaningful subset experiences bothersome tinnitus that interferes with concentration and mood ([Tunkel et al., 2014](#); [McCormack et al., 2016](#)). In employed middle-aged adults, tinnitus frequently emerges in a context of psychosocial stress, long working hours, sleep

disruption, occupational or recreational noise, cardiometabolic risk, and delayed help-seeking. Because these factors are temporally and biologically intertwined, simple causal narratives are unreliable and can mislead both patients and clinicians.

1.1 Motivating clinical observation

The framework is motivated by a common and instructive clinical trajectory, presented here as an anonymized composite rather than as identifiable patient data. A worker in the late 40s to early 50s first experiences unilateral tinnitus in one ear during a period of heavy workload. Reasoning intuitively, the person “rests” the affected ear by shifting phone calls and earphone use to the contralateral side; over roughly nine months the tinnitus resolves. About a year later, the previously unaffected ear develops sudden aural fullness and a “pop” during a stressful workday, followed within days to weeks by reduced audibility of soft sounds. Lacking awareness of the narrow treatment window for sudden sensorineural hearing loss (SSNHL), the person delays evaluation for roughly two months; by the time intratympanic and systemic steroid therapy is attempted, recovery is incomplete. This trajectory highlights two clinically actionable points that STARS (Stress, Tinnitus, and Audiometric Research Study) is built around: (i) stress-associated tinnitus is common and often self-limiting, but (ii) sudden or rapidly progressive hearing loss is a medical urgency in which delayed care—not stress per se—may drive irreversible impairment. Attributing acute hearing symptoms to “stress” without audiological and otolaryngological assessment is therefore hazardous.

1.2 Conservative premise and exposure naming

STARS begins from a conservative and testable premise: *perceived stress and work-related factors* may be *associated* with tinnitus and hearing outcomes, but public cross-sectional datasets are primarily suited to estimating associations, not to proving that stress directly causes tinnitus or sensorineural hearing loss. We adopt a strict naming principle throughout: because KNHANES and NHANES contain general perceived-stress, mood, and employment

items rather than validated job-strain instruments (e.g., the Job Content Questionnaire or Effort–Reward Imbalance scale), we never describe the exposure as measured “occupational stress.” It is labeled *perceived stress* (the psychological-distress/stress item) with *work-related factors* (occupation, employment status, and, where available, working hours and shift work) modeled as distinct covariates or effect modifiers. This distinction is central to the framework’s statistical claims and to its clinical safety posture.

1.3 A survey-weighted study reported against a prespecified plan

This is a survey-weighted research study whose *primary* hypothesis test—the perceived-stress analysis in KNHANES—is reported in full (Section 6), with supporting NHANES analyses. To limit analyst degrees of freedom (e.g., selecting cycles or covariate sets after seeing results), the design, variable definitions, survey cycles, adjustment sets, and evaluation plan were fixed *before* analysis, and the analysis code and committed outputs make every reported value reproducible. Three further components—cross-national validation, clinician-verified AI extraction, and a clinical extension—are *prospective*: their designs are specified here, but they carry no results and are deliberately kept subordinate to the empirical finding so as not to blur the article’s identity.

1.4 Aims and hypotheses

The framework has four aims, each with prespecified, falsifiable hypotheses. **Aim 1 is the empirical contribution of this article and is tested and reported here**; aims 2–4 are prospective and their designs are specified for a companion program of work.

1. **Association (development).** Estimate weighted associations between perceived stress and (a) tinnitus status and (b) audiometric hearing loss in KNHANES, adjusting for noise exposure and health covariates. *H1: A positive association between perceived stress and tinnitus persists after adjustment; the association with audiometric thresholds is weaker*

and may attenuate to non-significance after adjusting for noise and age.

2. **Transportability (external validation).** Test whether harmonized feature definitions and prediction models developed in KNHANES retain discrimination and calibration in NHANES. *H2: Discrimination degrades modestly but calibration can be restored by recalibration, supporting cross-national transportability of the tinnitus model more than the hearing-loss model.*
3. **Explainable, safe AI.** Evaluate whether an open medical language model can extract audiology-relevant fields from deidentified narratives at clinician-verified accuracy while a deterministic red-flag layer reliably routes SSNHL and neurologic warning signs to urgent referral. *H3: Schema-constrained extraction reaches high human-verified agreement with a low unsupported-output rate, and the safety layer achieves near-complete red-flag recall.*
4. **Open, shared benefit.** Provide an openly licensed, reproducible pipeline (code, data dictionary, model cards, prompts) that lowers the barrier for researchers to reproduce analyses and add local cohorts.

1.5 Contributions

The article’s contributions are ordered by what it demonstrates. *First* and primary: a survey-weighted, public-data result showing that perceived stress dissociates the tinnitus *symptom* from the audiometric *threshold*, reported without unsupported causal claims and reproducible end-to-end from committed code. *Second*: an open, reproducible scaffold—harmonized variable mappings, executable design-based estimators, model cards, and safety tests—that lets others regenerate every number and add local cohorts. The remaining contributions are *prospective* (design only): a prespecified KNHANES→NHANES external-validation plan (calibration, recalibration, clinical utility, subgroup fairness), and a clinically governed AI component in which schema-constrained language-model extraction is clinician-verified and a deterministic red-flag layer overrides probabilistic predictions for time-critical hearing

symptoms.

1.6 Scope and positioning

The spine of this AJA-targeted article is the empirical symptom-versus-threshold finding in KNHANES, with NHANES as directional support and the open scaffold as its reproducibility guarantee. The prospective components—cross-national validation of *research-only* risk-stratification models and a deterministic referral-safety layer for sudden hearing loss—are presented as a clearly bounded companion program, not as results of this study; any prediction model is explicitly a research risk-stratification tool, not a screening, diagnostic, or triage device, with intended use specified in Table S1. AI is an enabling research tool throughout, never a replacement for clinical judgment. Optional neurophysiological substudies (OpenNeuro tinnitus imaging/EEG) and audiogram-digitization of legacy records are noted only briefly as future directions and are out of scope here.

2 Related Work

2.1 Epidemiology of tinnitus, hearing loss, and stress

Population studies have repeatedly shown that tinnitus and hearing loss often coexist, but their relationships are heterogeneous. Analyses using KNHANES have estimated tinnitus prevalence and associated factors in Korea and demonstrated that nationally representative data can support clinically meaningful tinnitus research (Park et al., 2014; Joo et al., 2015). NHANES has similarly enabled U.S. studies of audiometry, occupational noise exposure, tinnitus, and mental-health correlates (Hoffman et al., 2017; Chakrabarty et al., 2024; Mahboubi et al., 2013). Reviews of tinnitus epidemiology emphasize wide prevalence estimates driven by heterogeneous case definitions, underscoring the need for harmonized variables when pooling or comparing across surveys (McCormack et al., 2016). Evidence linking psychological stress to tinnitus onset and severity is consistent but largely observational; longitudinal occupational

cohorts suggest stress is associated with incident tinnitus, yet residual confounding by noise, sleep, and cardiometabolic factors is difficult to exclude (Baigi et al., 2011).

2.2 What STARS adds over prior KNHANES tinnitus research

Because this study uses the KNHANES 2010–2012 otologic cycles, it must state precisely what is new relative to prior work on the same data. Earlier KNHANES analyses of this cycle range established tinnitus prevalence and its associated factors (Park et al., 2014) and its relationship to health-related quality of life (Joo et al., 2015), and related KNHANES work on these same cycles has examined tinnitus alongside psychological and mental-health correlates—most closely Lee et al. (2018), who reported higher perceived stress, melancholy, and suicidal ideation among premenopausal women with tinnitus. STARS is therefore *not* the first to link tinnitus with distress in KNHANES; its contribution is narrower and methodological. Specifically, STARS adds (i) a *prespecified, falsifiable* contrast between the tinnitus *symptom* and the audiometric *threshold* as the primary target (H1), rather than reporting a single tinnitus–distress association; (ii) an explicit DAG that separates *confounders* from *mediators* and reports both minimal-sufficient and mediator-adjusted models, so the estimate is interpretable rather than a kitchen-sink adjustment; (iii) a cross-national *directional-consistency* check in NHANES under a related distress construct; and (iv) a fully reproducible, open scaffold that regenerates every number from committed code and data mappings. The novelty is in the design discipline and reproducibility, not in the mere existence of a tinnitus–stress association.

2.3 Beyond the pure-tone average

Recent audiology work highlights the need to measure more than the better-ear pure-tone average. Studies of executive control and attention in individuals with tinnitus suggest that cognitive burden may be relevant to perceived functional impact, and speech-perception-in-noise and hearing-aid signal-management research shows that real-world listening difficulty

can persist even when threshold-based metrics appear near-normal. These directions motivate STARS to separate tinnitus presence from bothersome tinnitus and to analyze worse-ear and high-frequency thresholds and asymmetry, not better-ear PTA alone.

2.4 Clinical pathways and the treatment window

Clinical practice guidelines frame the endpoints STARS must respect. The clinical practice guideline for tinnitus distinguishes bothersome from non-bothersome tinnitus and prioritizes education, cognitive-behavioral strategies, and sound therapy over pharmacologic cures (Tunkel et al., 2014). Critically, the guideline for sudden hearing loss defines SSNHL as a medical urgency: prompt audiometric confirmation and early corticosteroid therapy within roughly two weeks offer the best chance of recovery, and delayed presentation is associated with worse outcomes (Chandrasekhar et al., 2019). This “golden-window” principle is the clinical rationale for the deterministic red-flag layer described in Section 5.

Prior work involving Hun Yi Park and colleagues is especially relevant for the clinical expansion of this study. Sound-localization findings in unilateral hearing-aid users show that the side of hearing impairment may influence spatial-hearing interpretation (Ha et al., 2022). Early-management research in traumatic benign paroxysmal positional vertigo, although not a tinnitus study, underscores the importance of rapid identification and pathway-based care for acute otologic and vestibular symptoms (Kim et al., 2022). Cochlear-implant research using a CT-derived basal turn–facial ridge angle illustrates how explainable anatomical variables can support prediction of residual hearing preservation (Kim et al., 2021). Together, these studies support a STARS design that separates screening, referral, treatment, and rehabilitation endpoints.

2.5 Open medical AI and reporting standards

Open medical AI models are increasingly available for research use. MedGemma is a collection of open models optimized for medical text and image comprehension, positioned as a starting

point for health-AI development rather than a validated diagnostic device (Google Research, 2025; Google Health AI Developer Foundations, 2026). For STARS, this implies constraining the model to schema extraction and evidence-linked explanations, with clinician verification and external validation before any clinical use. Methodologically, the framework adheres to established reporting standards for observational and prediction research—STROBE for observational analyses (von Elm et al., 2007) and the TRIPOD+AI statement for AI-based prediction models (Collins et al., 2024)—so that reported models are transparent, comparable, and auditable.

3 Public Dataset Selection

Dataset selection was governed by four requirements for the primary association analyses: (i) survey-weighted population representativeness; (ii) objective pure-tone audiometry; (iii) tinnitus and/or hearing-difficulty items; and (iv) covariates for noise exposure, psychosocial stress, and general health. Only KNHANES and NHANES jointly satisfy all four at population scale; other public resources are retained as optional secondary datasets. Table 4 summarizes the selection, and Table 5 maps target variables to each source.

3.1 Primary Development Dataset: KNHANES

KNHANES is selected as the primary development dataset because it is nationally representative for Korea and has previously supported tinnitus and hearing-loss analyses (Park et al., 2014; Joo et al., 2015). Its advantages include Korean population relevance, complex-survey weights, pure-tone audiometry in applicable cycles (otologic examination cycles 2010–2012), tinnitus and hearing-difficulty questionnaire items in selected cycles, health and behavioral covariates, and psychosocial or stress-related self-report variables (e.g., perceived-stress and depressed-mood items). It also provides a realistic bridge to future Korean hospital cohorts, including the planned Ajou University Hospital extension. Because item wording and au-

diometry availability differ across cycles, the analysis prespecifies a per-cycle codebook review and restricts pooling to cycles with comparable definitions.

The limitations are equally important. KNHANES is not a workplace cohort, and the audiometric analyses are cross-sectional. It can therefore evaluate *associations* among perceived stress, work-related variables, tinnitus, and audiometric outcomes, but cannot by itself establish that occupational stress *caused* tinnitus or hearing loss. Treatment timing and recovery after sudden hearing loss are not captured.

3.2 External-Validation Dataset: NHANES

NHANES is selected as the primary external-validation dataset because it provides openly downloadable U.S. public-use files, pure-tone audiometry and tympanometry in selected cycles (e.g., 2011–2012, 2015–2016, 2017–March 2020), occupational and recreational noise-exposure items, tinnitus and hearing questionnaires in selected cycles, and extensive demographic and health covariates. NHANES enables testing whether feature definitions and risk models derived from KNHANES retain discrimination and calibration across population, language, health-system, and occupational contexts. As with KNHANES, complex-survey weights, strata, and primary sampling units must be honored, and cycle-specific age eligibility for the audiometry module must be respected.

3.3 Optional Secondary Datasets

Open audiology repositories may be used for targeted secondary experiments: the Oldenburg Hearing Health Repository and Zenodo auditory-test datasets for audiometric model benchmarking and audiogram-based clustering; OpenNeuro tinnitus neuroimaging and fNIRS datasets for an optional neurophysiological substudy of tinnitus subtypes; and Mendeley/Zenodo tinnitus EEG datasets for explainable-biomarker exploration. These resources are not primary because they typically lack the joint combination of population weights, tinnitus items, audiometry, occupational factors, and general-health covariates required for the main

survey-weighted analyses. Any such dataset is used only after verifying its license, consent basis, and permission for secondary research use.

3.4 Prospective Clinical Extension (Ajou University Hospital)

A prospective otolaryngology/audiology cohort is specified to address what public cross-sectional data cannot: treatment timing, SSNHL recovery trajectories, tinnitus burden over time, and hearing-aid rehabilitation. This cohort contributes deidentified audiograms and clinical narratives for clinical external validation and provides the substrate for the AI extraction and safety evaluations. Raw clinical data are not redistributed; only derived, deidentified, or synthetic artifacts are shared through the open repository, subject to institutional review board (IRB) approval and data-governance review.

4 Methods (Prespecified)

The primary observational analysis is reported following STROBE (von Elm et al., 2007); the prospective prediction and clinical components follow TRIPOD+AI (Collins et al., 2024) and SPIRIT, respectively. Completed reporting checklists accompany the repository. To limit analyst degrees of freedom, all design choices below were fixed prior to real-data analysis.

4.1 Study Design

The work has three stages, summarized in the conceptual framework of Figure 1. **Stage 1** analyzes KNHANES as the primary public-data development cohort (association estimation and research-model development). **Stage 2** evaluates cross-national transportability in NHANES (geographic external validation). **Stage 3** prepares a prospective clinical extension at Ajou University Hospital for clinical external validation, treatment-timing analysis in SSNHL, and evaluation of the AI extraction and safety components. Stages 1–2 use only deidentified public-use files; Stage 3 proceeds only under IRB approval.

4.2 Fixed Datasets, Cycles, and Eligibility

To eliminate analyst degrees of freedom, the survey cycles, files, ages, frequencies, and exclusions are fixed in advance (Table 4). **KNHANES:** the otologic-examination cycles carrying pure-tone audiometry and tinnitus items (KNHANES V, 2010–2012) form the primary development window; cycles are pooled only after confirming identical item wording and audiometry protocols. **NHANES:** the audiometry-module cycles 2011–2012, 2015–2016, and 2017–2018 form the external-validation window. For both, eligible records are adults aged 40–69 years (the band jointly covered by the audiometry modules and our target of mid-to-late-career workers), with valid examination weights and a non-missing value for the outcome under analysis. Pure-tone thresholds are analyzed at 0.5, 1, 2, 3, 4, 6, and 8 kHz per ear; conductive patterns are flagged and, where tympanometry/air–bone data permit, excluded in a sensitivity analysis. We prespecify expected raw and analytic sample sizes per window (reported in the repository data dictionary) and the exact set of predictors available in *both* surveys; any predictor available in only one survey is excluded from the common transportable model and used only in country-specific extended models.

4.3 Exposure: Perceived Stress and Work-Related Factors

Consistent with the naming principle (Introduction), the primary exposure is *perceived stress*, operationalized from each survey’s general stress/psychological-distress item, and *work-related factors* (occupation category, employment status, and—where a cycle provides them—working hours and shift work) are modeled as separate covariates or effect modifiers, never as a proxy for validated job strain. Because item wording, response categories, and cultural response styles differ across surveys, Table 6 records, for every exposure and covariate, the original question, response scale, harmonization rule, and a measurement-comparability rating (high/medium/low). Variables rated *low* comparability (typically the stress and some psychosocial items) are (a) excluded from the common transportable model, (b) entered only in country-specific models, and (c) examined in a formal measurement-invariance check

(configural/metric where a latent construct is defined). We do not pool a variable across countries unless at least metric-level comparability is plausible.

4.4 Causal Structure and Adjustment Sets

Figure 2 presents the directed acyclic graph (DAG) encoding our assumptions. Age, sex, education, occupation/employment, and noise exposure are treated as confounders of the perceived-stress→outcome relationships and constitute the **minimal sufficient adjustment set** for the primary analysis. Depressed mood and sleep are treated as *potential mediators* of the stress→tinnitus path (and possibly colliders when conditioning opens non-causal paths); conditioning on them can constitute overadjustment. Accordingly:

- **Primary model** adjusts for the minimal sufficient set only (age, sex, education, occupation/employment, noise).
- **Extended model** additionally includes depressed mood, reported separately and interpreted as a mediator-adjusted (not confounder-adjusted) estimate; a **fully adjusted** sensitivity model further adds socioeconomic status (education, income) and, where available, smoking and cardiometabolic comorbidity, to address residual confounding by conventional covariates.
- **Mediation** is summarized by the mediator-adjusted estimate together with an exploratory difference-method proportion (the attenuation of the perceived-stress log-odds after adjusting for depressed mood); we do not claim a formal counterfactual (direct/indirect) decomposition, because cross-sectional data and logistic non-collapsibility make such estimates unreliable.
- **Collider check** enumerates conditioning sets that could induce collider bias and excludes them from the primary specification.

Because restricting to employed participants can induce a healthy-worker effect (and a selection collider), the primary population is *all eligible adults*, with an *employed-restricted*

analysis reported as a prespecified secondary contrast rather than the main result.

4.5 Outcomes

Endpoints are defined a priori and separated:

- **Tinnitus presence** (any tinnitus in the past 12 months) and **bothersome tinnitus** (tinnitus causing annoyance/sleep or concentration difficulty) are modeled as *distinct* endpoints, since their determinants differ.
- **Hearing loss** is quantified by the speech-frequency pure-tone average (PTA_{0.5,1,2,4}) for *both the better ear and the worse ear*, plus a **per-ear** (ear-level, mixed-effects) analysis, because a better-ear summary can mask the unilateral/asymmetric presentations most relevant to SSNHL. The primary threshold is PTA > 25 dB HL (the WHO grade-1 cutoff ([World Health Organization, 2021](#))); > 20 dB HL and high-frequency PTA_{3,4,6} are prespecified sensitivity definitions, with the 25 dB HL choice justified as the established clinical cutpoint.
- **Asymmetric hearing loss** is defined as an interaural difference ≥ 15 dB at ≥ 2 adjacent frequencies, consistent with asymmetry criteria used in sudden/asymmetric hearing-loss guidance ([Chandrasekhar et al., 2019](#)).
- **Secondary:** self-reported hearing difficulty, hearing-aid use, and quality-of-life correlates.

Noise exposure is coded, where cycles permit, not merely as a binary but with duration, intensity/source (occupational, recreational, firearm), and hearing-protection use, so residual confounding by noise is minimized.

4.6 Statistical Analysis

All public-data analyses use survey-aware methods (Taylor-linearization variance estimation honoring weights, strata, and PSUs). The analytic age band (40–69) is estimated as a

subpopulation (domain) that retains the full survey design rather than by subsetting the data before variance estimation, and single-PSU (“lonely”) strata are handled by the standard conservative centering option (centering the lone unit at the grand mean) rather than by dropping the stratum term, which would under-estimate variance. Point estimates and design-based standard errors are reproducible both from the committed Python estimators (unit-tested in `code/src/test_nhanes_analysis.py`) and, independently, from the R `survey` package via `code/src/reproduce_survey.R` (`svydesign + subset` domain estimation with `survey.lonely.psu="adjust"`). In this cohort, domain estimation and lonely-PSU centering left the headline confidence intervals essentially unchanged (e.g., the tinnitus OR remained 1.42, 95% CI 1.26–1.60), indicating the estimates are not sensitive to these design choices. Multiplicity across the prespecified secondary contrasts is controlled by Benjamini–Hochberg false-discovery-rate-adjusted q -values (reported with the sensitivity models), with the two primary endpoints reserved as the confirmatory H1 test. Descriptive analyses estimate weighted prevalence with 95% confidence intervals. Association analyses use weighted logistic, ordinal, or linear regression by outcome type. The **primary specification adjusts for the minimal sufficient set**; the extended (mediator-adjusted) model is secondary. Results are reported as associations (odds ratios or mean differences) with 95% CIs, explicitly not as causal effects. Multiplicity is handled by prespecifying primary hypotheses (Introduction) and controlling the false-discovery rate across secondary contrasts.

Missing data. Missingness is described by variable and mechanism. The KNHANES tinnitus/audiometry outcomes come from the otology (ENT) examination, a ≥ 40 sub-sample; because the KDCA does not release an ENT-examination-specific weight for these cycles, we use the distributed interview-plus-examination weight (`wt_itvex`), pooled across 2010–2012 by dividing by the number of years per KDCA guidance, and treat any residual calibration mismatch for the ENT sub-sample as a weighting limitation (Discussion). The primary reported analysis is *complete-case* (design-based estimation on records with the outcome and

model covariates observed); we report this transparently as the executed analysis. Multiple imputation by chained equations (MICE) under a missing-at-random assumption, with design variables in the imputation model and pooling by Rubin’s rules, is a prespecified sensitivity analysis. This is a deliberate, documented deviation from an earlier plan that named MICE as primary: because item missingness in the analytic variables is low and the design-based estimators handle it directly, complete-case is the primary estimator and MICE is reported as a robustness check rather than the headline. In external validation, records missing a common predictor are handled by the same prespecified rule in both surveys.

Sample size and precision. Sample sizes are fixed, so we frame precision rather than post-hoc power: with several thousand audiometry participants per window, weighted prevalence is estimable to within a few percentage points and moderate associations ($OR \approx 1.3$ – 1.5) are detectable at conventional error rates. Exact expected analytic N per window is fixed in the repository before analysis.

4.7 Research-Only Prediction Models and Intended Use

Two targets—tinnitus presence and audiometric hearing loss—are developed as *research risk-stratification models*, not screening, diagnostic, or triage devices. Their intended use, target user, prediction time, inputs, decision, and the costs of false positives/negatives and contraindicated uses are specified in Table S1. We first fit transparent survey-weighted logistic baselines with prespecified predictors, then gradient-boosted trees as a comparator; the simpler model is preferred unless the complex model yields a clinically meaningful, calibrated improvement. Development uses KNHANES with repeated stratified cross-validation and optimism correction. All targets, predictors, and metrics are frozen before external validation (Table S3). To verify wiring, metric computation, and reporting format only, the repository ships a synthetic-data demonstration whose outputs (e.g., a tinnitus-target AUROC ≈ 0.64 on randomly generated data) carry *no* epidemiological meaning and are deliberately excluded

from the Results and abstract; substituting real files routes the identical code path to the design-based estimates reported above. SHAP values are reported for interpretability, with the explicit caveat that SHAP does not guarantee causal or even stable attributions when predictors are correlated; importance is interpreted descriptively, not causally.

4.8 External Validation Across Survey Designs

Transporting a KHNHANES model to NHANES confronts differences in weighting, population prevalence, age range, category coding, and health-system context. We therefore prespecify:

- **Unweighted vs. weighted performance** reported separately—unweighted for individual-level prediction, survey-weighted for population-level performance.
- **Calibration-in-the-large and calibration slope reported separately**, before and after recalibration (intercept-only, then intercept+slope update of the frozen model).
- **Common-support analysis** restricting comparison to the overlapping covariate region of the two surveys.
- **Case-mix standardization** to distinguish genuine model transportability from case-mix differences in prevalence/age.
- **A reduced common model** (predictors available and comparable in both surveys) versus **country-specific extended models**, so that limited measurement invariance does not masquerade as poor transportability.

Fairness and subgroup evaluation. Discrimination and calibration are reported within prespecified subgroups: sex, age band, employment status, noise-exposure status, unilateral/asymmetric hearing, *socioeconomic status* (education/income), and *race/ethnicity* (NHANES, where collected). Subgroup calibration and error gaps are first-class results, not afterthoughts.

4.9 Sensitivity Analyses

Prespecified sensitivity analyses test alternative hearing-loss definitions (25 vs. 20 dB HL; high-frequency PTA; worse-ear and per-ear), age restrictions, exclusion of conductive pat- terns when tympanometry permits, alternative stress operationalizations, mediator- vs. confounder-treatment of depressed mood/sleep, all-adult vs. employed-restricted populations, and stratification by noise exposure, sex, and asymmetry.

4.10 Ethics and Governance

Stages 1–2 use publicly available, deidentified survey data and do not constitute human- subjects research requiring new consent; KNHANES and NHANES were approved by their respective review boards, and their use follows each provider’s data-use terms. The Stage- 3 clinical extension will obtain IRB approval at Ajou University Hospital, with a data- management plan covering deidentification, access control, and secondary-use consent. No identifiable patient data are placed in the public repository; only derived, deidentified, or synthetic artifacts are shared.

5 Prospective Companion Component: Open AI Sys- tem and Referral Safety

Scope note. This section specifies a **prospective** companion component; it carries **no results** and is not part of the empirical contribution of this article. It is included so that the design, safety governance, and evaluation plan are fixed in advance and can be reported and executed in a dedicated follow-up under IRB approval. Readers seeking the study’s empirical findings should see Section 6. The deterministic referral-safety layer and the schema-constrained extraction harness are developed, benchmarked, and reported *with results* in a separate companion paper; this section fixes only their role within STARS.

5.1 Motivation

A major barrier to audiology AI is the scarcity of harmonized, reusable clinical features. STARS addresses this by publishing a transparent feature schema, reproducible public-data code, and a clinician-verification protocol. This supports a broadly shared-benefit research model: investigators can reuse the pipeline, add local cohorts, and compare models without exposing identifiable patient data.

5.2 Model Roles and Prohibitions

The AI system has four restricted roles, each with an explicit safety constraint (Table S2): (1) it harmonizes variable names and labels across public datasets; (2) it trains transparent baseline models for association and calibrated risk stratification; (3) in the clinical extension, an open medical language model (e.g., MedGemma) extracts structured fields from deidentified notes using a fixed JSON schema; and (4) it drafts explanations that cite only approved study documents or clinical guidelines. The system is explicitly prohibited from diagnosing an individual patient, assigning a definitive etiology, recommending or dosing steroids, or delaying urgent referral. These prohibitions are enforced in code and documented in a model card.

5.3 Deterministic Red-Flag Safety Layer

Before any probabilistic model output is surfaced, a deterministic rule layer screens for red flags derived from the SSNHL and tinnitus guidelines (Chandrasekhar et al., 2019; Tunkel et al., 2014): sudden or rapidly progressive hearing loss (especially unilateral), aural fullness with a discrete onset, single-sided tinnitus with asymmetry, vertigo, or focal neurologic signs. Any trigger overrides the model and returns an “urgent audiology/otolaryngology referral” message. This operationalizes the “golden-window” lesson from the motivating vignette: the framework must never let a probabilistic “low-risk, likely stress” estimate suppress timely

referral.

Evaluation design. The safety layer’s evaluation is prespecified in detail (Table S4) rather than asserted as a target recall alone. A vignette set of ≥ 200 cases mixes real deidentified, expert-authored, and synthetic cases in a fixed ratio; each is labeled by at least two otolaryngology/audiology experts with inter-rater agreement (κ) reported before adjudication. We report sensitivity (red-flag recall) with 95% CIs as the primary metric and specificity/over-referral as secondary, stratified by laterality, onset speed, and vertigo/neurologic signs, with explicit robustness tests for negation, ambiguous temporal expressions, and colloquial patient phrasing. Every missed red flag is audited as a critical failure. Crucially, we state that a sensitivity of 1.0 on a finite test set is a target, *not* a guarantee of zero misses in deployment; the layer is a safeguard layered on top of, never a replacement for, clinical assessment.

5.4 Schema-Constrained Extraction and Governance

LLM extraction is constrained to a fixed schema with typed fields and controlled vocabularies; free-text generation is disallowed for structured outputs. Each extracted field carries a source span and a confidence flag, and every field is clinician-verified before entering analysis. Prompts, schema versions, model weights/version, temperature, and random seeds are logged. We define the *confidence flag* explicitly as a rule-based reliability indicator (presence/agreement of a supporting source span), not the language model’s own token probability, to avoid conflating self-reported confidence with correctness.

Extraction evaluation. Because “field accuracy” alone is inadequate for structured extraction, we report per field: exact-match rate; precision, recall, and F1; omission rate; evidence-span agreement; and dedicated accuracy for temporal expressions, laterality, onset speed, and symptom duration. We further report a clinically significant error rate (errors that could change referral), document-level exact-match rate, run-to-run consistency across

repeated generations, clinician correction time, and performance by model version. Targets and thresholds are fixed before evaluation.

5.5 Robustness and Trustworthiness

Trustworthiness is pursued through eight design choices: (1) public-data baselines; (2) country-level external validation; (3) survey-weighted epidemiology; (4) interpretable models before complex models; (5) locked feature and outcome definitions; (6) clinician review of LLM-extracted variables; (7) subgroup calibration and error analysis; and (8) full run provenance (dataset versions, variable mappings, hyperparameters, prompts, seeds). Reproducibility is supported by pinned dependencies and deterministic seeds.

5.6 Open-Science Contribution

The repository includes executable analysis templates, a data dictionary, model cards, prompt templates, a red-flag test suite, and a completed reporting checklist. Because KNHANES files cannot be redistributed without official access procedures, the code specifies directory expectations and variable-mapping scripts rather than repackaging raw data. NHANES download helpers retrieve public files directly from CDC endpoints where available. A synthetic-data generator lets others exercise the full pipeline end-to-end without any restricted data, lowering the barrier to reuse and review.

6 Results

This section reports the *primary* perceived-stress analyses in the KNHANES development cohort, followed by the *supporting* NHANES analyses. KNHANES is the primary test of the aims because, unlike NHANES, it carries a general perceived-stress item (the NHANES psychological-distress proxy is the PHQ-9 depression screen, which the study DAG treats as a *mediator*). All public-data estimates are design-based (survey-weighted, Taylor-linearized)

and are associations, not causal effects.

6.1 Primary results (KNHANES development cohort)

The primary analyses use real KNHANES microdata for Korean adults aged 40–69, pooling the otologic-examination cycles 2010–2012 (KNHANES V). The perceived-stress and covariate items (health-interview file) were merged on the person identifier to the separate ENT (otolaryngology) examination file that carries pure-tone audiometry, the tinnitus item, and occupational-noise exposure; the survey weight (`wt_itvex`) was pooled across the three years and Taylor-linearized. The analysis is produced by `code/src/knhanes_analysis.py` with `config/knhanes_mapping.yaml` (the same validated design-based estimators as the NHANES pipeline), and the committed `outputs/knhanes_results.json` reproduces every value below. Of the pooled otologic-examination records for adults 40–69 (analytic domain $n = 10,467$), the complete-case models retained 9,142–9,532 participants depending on the outcome, with the remainder excluded for an item-missing outcome or covariate (STROBE participant flow; per-variable non-missing counts are printed by the pipeline and recorded in the repository).

Table 1: **Real KNHANES results** (Korean adults 40–69). Primary development cohort with the perceived-stress exposure. Prevalence is design-based (survey-weighted, Taylor-linearized); associations are mutually adjusted design-based odds ratios and are *not* causal. Generated by `code/src/knhanes_analysis.py`. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Tinnitus	Hearing loss (>25 dB HL)
Prevalence (95% CI)	21.1% (19.9–22.3)	15.4% (14.5–16.4)
<i>Adjusted associations, OR (95% CI)</i>		
Perceived stress (high)	1.42 (1.26–1.60)***	1.18 (0.99–1.41)
Occupational noise	1.37 (1.15–1.63)***	1.72 (1.44–2.06)***
Age (per year)	1.03 (1.02–1.04)***	1.13 (1.12–1.14)***
Female sex	1.08 (0.96–1.22)	0.47 (0.40–0.55)***

Prevalence. Survey-weighted prevalence among adults 40–69 was 21.1% (95% CI 19.9–22.3; $n = 9,657$) for any tinnitus, 7.0% (6.3–7.7) for bothersome tinnitus (moderate-or-severe

annoyance on the KHNHANES 3-level item, $\approx 33\%$ of those with tinnitus), 15.4% (14.5–16.4; $n = 9,325$) for better-ear speech-frequency hearing loss (>25 dB HL), and 24.3% (23.2–25.3; $n = 9,751$) for high perceived stress.

Primary association (H1). In the mutually adjusted, minimal-sufficient model (perceived stress, occupational noise, age, sex), high perceived stress was associated with *tinnitus* (OR 1.42, 1.26–1.60), alongside occupational noise (OR 1.37, 1.15–1.63) and age (OR 1.03 per year), with no significant sex difference. Its association with *audiometric hearing loss* was weaker and did not reach significance (OR 1.18, 0.99–1.41); there, occupational noise (OR 1.72, 1.44–2.06), age (OR 1.13 per year), and male sex (female sex OR 0.47, 0.40–0.55) dominated. These are design-based associations, not causal effects.

Interpretation (H1 supported). This is precisely the prespecified pattern of H1, now demonstrated in the primary cohort with the *actual* perceived-stress exposure: perceived stress tracks the tinnitus *symptom* (OR 1.42, $p < 0.001$) but attenuates against the audiometric *threshold* (OR 1.18, $p = 0.07$) once occupational noise and age are accounted for. The symptom-versus-threshold dissociation seen preliminarily in NHANES (with a distress proxy) thus holds for the genuine stress exposure in KHNHANES, consistent with a stress–tinnitus association that is not primarily explained by measurable cochlear threshold shift.

Extended and sensitivity models. We tested whether the primary stress–tinnitus association is an artifact of under-adjustment or of reverse causation using prespecified secondary specifications that isolate the perceived-stress odds ratio (Table 2). The association *survived* adjustment for the mediator depressed mood (OR 1.29, 1.13–1.47) and, importantly, adjustment for *audiometric hearing loss* (OR 1.44, 1.27–1.64), one of the strongest tinnitus correlates: stress tracks tinnitus over and above measurable threshold shift and mood. A *fully adjusted* model additionally including socioeconomic status (education, income) and—where the codebook variables are supplied—smoking and cardiometabolic comorbidity is reported

in the same table (Table 2) to address residual confounding by conventional covariates. In an exploratory difference-method decomposition, adjustment for depressed mood attenuated the perceived-stress log-odds for tinnitus by $\approx 27\%$ (OR 1.42 $\rightarrow$ 1.29)—an estimate of the share statistically consistent with mediation through mood, read cautiously given the cross-sectional design and logistic non-collapsibility. For hearing loss, the null better-ear estimate persisted after mediator adjustment (OR 1.11, 0.92–1.34), but the dissociation was *relative, not absolute*: perceived stress showed a weak association with *worse-ear* speech-frequency hearing loss (OR 1.15, 1.00–1.31)—the ear most relevant to unilateral/asymmetric disease—while the *high-frequency* (3/4/6 kHz) definition remained null (OR 1.07, 0.93–1.23).

Reverse-causation sensitivity and multiplicity. Because bothersome tinnitus is itself a stressor, we contrasted bothersome and non-bothersome cases. Perceived stress was associated with bothersome tinnitus most strongly (OR 1.79, 1.46–2.21) but *remained* associated with non-bothersome tinnitus versus no tinnitus (OR 1.24, 1.05–1.45), a gradient that argues against a purely reverse-causal explanation while remaining consistent with a feedback contribution (Discussion). Across the seven secondary contrasts we control the false-discovery rate (Benjamini–Hochberg); the mediator-adjusted, hearing-loss-adjusted, bothersome, and non-bothersome tinnitus associations remain significant ($q < 0.02$), whereas the borderline worse-ear association does *not* survive adjustment ($q \approx 0.07$) and should be read cautiously. The two primary endpoints (tinnitus and better-ear hearing loss) constitute the confirmatory H1 test and are not part of this secondary family.

Audiometric protocol and conductive contamination. The KNHANES ENT examination provides pure-tone *air-conduction* thresholds per ear; the better-ear speech-frequency PTA is the primary hearing-loss definition, with the worse-ear and high-frequency definitions above as prespecified sensitivity analyses. Because air-conduction thresholds alone cannot separate conductive from sensorineural loss, any conductive component is a source of outcome misclassification; the availability of tympanometry/bone-conduction data to exclude conduc-

Table 2: **Extended and sensitivity models (KNHANES, adults 40–69).** Perceived-stress odds ratio (95% CI) across prespecified specifications, isolating the robustness of the primary exposure. Rows add the mediator (depressed mood) or audiometric hearing loss to the minimal-sufficient set (perceived stress, occupational noise, age, sex), and vary the hearing-loss outcome definition (better-ear, worse-ear, high-frequency). All are design-based associations, not causal effects; mediator-adjusted rows are interpreted as mediator- (not confounder-) adjusted. Bothering tinnitus (moderate/severe annoyance) is modeled versus all others; non-bothering tinnitus (present but not annoying) versus no tinnitus, excluding bothersome cases. Secondary contrasts are Benjamini–Hochberg false-discovery-rate controlled (reported in text). Generated by `code/src/knhanes_analysis.py`. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Model specification	Perceived stress, OR (95% CI)	<i>N</i>
Tinnitus — minimal-sufficient (primary)	1.42 (1.26–1.60)***	9504
+ depressed mood (mediator-adjusted)	1.29 (1.13–1.47)***	9440
+ audiometric hearing loss	1.44 (1.27–1.64)***	9142
+ SES/comorbidity (fully adjusted)	1.29 (1.13–1.48)***	9307
Hearing loss, better ear — minimal-sufficient	1.18 (0.99–1.41)	9176
+ depressed mood (mediator-adjusted)	1.11 (0.92–1.34)	9131
Hearing loss, worse ear — minimal-sufficient	1.15 (1.00–1.31)*	9176
Hearing loss, high-freq (3/4/6 kHz) — minimal-sufficient	1.07 (0.93–1.23)	9176
Bothering tinnitus — minimal-sufficient	1.79 (1.45–2.21)***	9532
Non-bothering tinnitus vs none (reverse-causation sensitivity)	1.24 (1.05–1.45)*	8781

tive patterns in these cycles is confirmed against the KNHANES user guide, and where such data are absent, conductive contamination is treated as a limitation (Discussion) rather than silently ignored.

Further prospective analyses. The employed-restricted (healthy-worker) contrast, the KNHANES→NHANES external validation (calibration, recalibration, common support, case-mix standardization; H2), and subgroup fairness (SES; race/ethnicity in NHANES) are prospective components that follow the prespecified plan (Methods and Section 5); they carry no results here.

6.2 Supporting results (NHANES): directional consistency

Using the tested, design-based pipeline (`code/src/nhanes_analysis.py`; estimators unit-tested in `code/src/test_nhanes_analysis.py`), we analyzed real NHANES public-use files for U.S. adults aged 40–69, pooling the 2011–2012, 2015–2016, and 2017–2018 cycles (Table 3).

Table 3: **Real NHANES results** (U.S. adults 40–69; pooled 2011–2012, 2015–2016, 2017–2018). Prevalence is design-based (survey-weighted, Taylor-linearized); associations are odds ratios (OR) from design-weighted logistic regression, mutually adjusted, and are *not* causal. Depressed mood is a PHQ-9 screen used here as the available psychological-distress proxy (NHANES lacks a validated perceived-stress instrument); per the study DAG it is a mediator, so its OR is a mediator-level association. Generated by `code/src/nhanes_analysis.py`. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Tinnitus	Hearing loss (>25 dB HL)
Prevalence (95% CI)	28.5% (26.5–30.4)	12.0% (10.4–13.6)
<i>Adjusted associations, OR (95% CI)</i>		
Occupational noise	1.59 (1.32–1.92)***	1.22 (0.94–1.58)
Depressed mood (PHQ-9≥10)	1.61 (1.23–2.11)***	2.29 (1.56–3.35)***
Age (per year)	1.00 (0.98–1.01)	1.11 (1.09–1.13)***
Female sex	1.61 (1.35–1.91)***	0.42 (0.29–0.61)***
Analytic N (associations)	4603	4463
Depression (PHQ-9≥10) prevalence: 8.4% (7.5–9.4), $N = 7293$		

Prevalence. Survey-weighted prevalence was 28.5% (95% CI 26.5–30.4; $n = 5,354$) for any tinnitus in the past 12 months, 12.0% (10.4–13.6; $n = 4,808$) for better-ear speech-frequency hearing loss (>25 dB HL), and 8.4% (7.5–9.4; $n = 7,293$) for a positive PHQ-9 depression screen.

Associations. In mutually adjusted design-based logistic models (adjusting for occupational noise, depressed mood, age, and sex), *tinnitus* was associated with occupational-noise exposure (OR 1.59, 1.32–1.92), depressed mood (OR 1.61, 1.23–2.11), and female sex (OR 1.61, 1.35–1.91), but not with age (OR 1.00, 0.98–1.01). *Hearing loss* was associated with age (OR 1.11 per year, 1.09–1.13) and depressed mood (OR 2.29, 1.56–3.35), inversely with female sex (OR 0.42, 0.29–0.61), and only weakly and non-significantly with occupational noise (OR 1.22, 0.94–1.58). These are associations, not causal effects.

Interpretation (a priori pattern). The pattern is coherent with the framework’s hypotheses. The psychological-distress proxy is associated with both outcomes, but more than doubles the odds of *audiometric* hearing loss while showing a smaller association with *tinnitus*; age dominates hearing loss but is unrelated to tinnitus in this band; and the sex pattern (higher tinnitus, lower hearing loss in women) matches known epidemiology. Because the proxy is a mediator in the DAG, these are mediator-level associations, and NHANES lacks the perceived-stress exposure; they therefore motivate—but do not substitute for—the primary KNHANES stress analysis. All values are reproduced by the committed results file and can be regenerated end-to-end from the CDC files.

6.3 Planned AI-component reporting

LLM-extracted fields will be reported with the per-field metric suite of Section 5 (exact match; precision/recall/F1; omission; span agreement; temporal/laterality/onset/duration accuracy; document-level match; run consistency; clinically significant error rate), plus clinician correction time and per-version performance. The red-flag layer will be reported

per Table S4 (dual-expert-labeled vignettes; sensitivity with 95% CI; specificity/over-referral; subgroup and robustness), with any missed red flag audited as a critical failure and with the explicit caveat that finite-set perfection does not guarantee zero deployment misses. Model cards accompany each released model, documenting intended use (Table S1), data provenance, subgroup metrics, and prohibitions.

7 Discussion

STARS reports one empirical finding and builds a prespecified companion program around it. The finding is a symptom-versus-threshold dissociation: in representative Korean adults, perceived stress is associated with the tinnitus symptom but not, after adjustment for occupational noise and age, with the audiometric threshold. The surrounding commitments are methodological rather than empirical: KNHANES is the development cohort and NHANES a directionally consistent support; AI is framed as a reproducible research accelerator bounded by a deterministic safety layer, not a diagnostic authority; and reproducibility is operationalized through open code, a data dictionary, model cards, safety tests, and a synthetic-data path.

7.1 Interpretation of the primary finding

The scientific claim is *not* that workplace stress damages hearing. Rather, the primary KNHANES analysis shows that *perceived stress* is associated with tinnitus but not, after the minimal sufficient adjustment, with the audiometric threshold: the stress–tinnitus association (OR 1.42) is robust to adjustment for occupational noise and age, whereas the stress–threshold association attenuates to non-significance (OR 1.18, $p = 0.07$) as noise and age dominate. This asymmetry, prespecified as H1, has a clear reading: perceived stress relates to tinnitus perception and burden—a partly central, attention-modulated phenomenon—more than to cochlear thresholds. Because depressed mood and sleep are modeled as mediators, the

primary estimate avoids overadjusting away a genuine indirect pathway; the prespecified extended model shows the association is largely independent of mood (mediator-adjusted OR 1.29) and, tellingly, of measurable hearing loss (hearing-loss-adjusted OR 1.44), so it is not an artifact of under-adjustment for the strongest tinnitus correlate. The dissociation is, however, *relative rather than absolute*: while the better-ear and high-frequency threshold associations are null, perceived stress is weakly associated with the *worse-ear* threshold (OR 1.15, 1.00–1.31), the ear most relevant to asymmetric disease—a signal we do not over-read, as it does not survive false-discovery-rate adjustment across the secondary contrasts ($q \approx 0.07$). On magnitude, an odds ratio of 1.42 is a modest but population-relevant increase in the odds of tinnitus at a stress prevalence of roughly one in four adults; we interpret it as a meaningful association rather than a large effect. We deliberately avoid dichotomizing the threshold result around $p = 0.05$: OR 1.18 with a 95% CI of 0.99–1.41 ($p = 0.07$) is best read as a *weak, attenuated* association whose confidence interval barely includes the null, not as evidence of “no association”; the contrast with the tinnitus estimate is one of degree and precision, not a categorical significant-versus-null split.

7.2 Directionality and reverse causation

The single most important threat to interpreting the stress–tinnitus association is *reverse causation*, and it deserves first-class treatment rather than a line in a limitations list. A cross-sectional odds ratio cannot distinguish “perceived stress is associated with having tinnitus” from “having tinnitus generates perceived stress.” The second direction is at least as plausible: bothersome tinnitus is itself a well-documented stressor that disrupts sleep, concentration, and mood, so a positive cross-sectional association is expected under a purely tinnitus→stress mechanism. Our design cannot resolve this, and we therefore make no claim that stress temporally precedes or produces tinnitus. Two considerations bound—without eliminating—the concern. First, our prespecified *sensitivity analysis restricted to non-bothersome tinnitus* versus no tinnitus is informative (Table 2): because non-bothersome tinnitus exerts weaker

feedback pressure on distress, the fact that the stress association *persists* there (OR 1.24, 1.05–1.45; $q < 0.02$ after false-discovery-rate adjustment) is harder to explain by reverse causation alone. At the same time, the association is strongest for bothersome tinnitus (OR 1.79), and this monotone gradient from no tinnitus to non-bothersome to bothersome is exactly what a reverse/feedback contribution would also produce. The honest reading is therefore mixed: the persistence in non-bothersome cases argues against a *purely* reverse-causal account, while the gradient is consistent with reverse causation contributing to the bothersome end. Second, the symptom-versus-threshold dissociation is more consistent with a central/attentional link than with stress-induced cochlear injury, but it too is direction-agnostic. Resolving temporality requires the prospective cohort (Stage 3), not these data.

7.3 Measurement validity of the exposure and outcome

Both the exposure and the primary outcome are *single self-report items*, and this bounds every estimate above. Perceived stress is one general item (BP1), not a validated instrument such as the PSS-10, the Job Content Questionnaire, or the Effort–Reward Imbalance scale; it captures a global sense of stress, not measured occupational stress, and we name it accordingly throughout. Tinnitus is likewise one item (past-12-month tinnitus, with a 3-level annoyance follow-up for the bothersome definition), with no validated tinnitus measure (THI/TFI) and no audiometric tinnitus confirmation. Crucially, if this measurement error is *non-differential* (independent of the other variable)—the usual expectation for general survey items—it biases the observed association *toward the null*, so the true stress–tinnitus association may be *larger* than the observed OR 1.42, and the attenuated stress–threshold association may be an under- rather than over-statement. We therefore interpret magnitudes cautiously and treat the single-item exposure as a floor on, not a ceiling of, the underlying relationship.

7.4 NHANES: directional consistency, not external validation

NHANES here provides *directional consistency*, not external validation of the stress exposure. NHANES carries no perceived-stress item; its available distress measure is a PHQ-9 depression screen, which the study DAG treats as a *mediator*—a related but distinct construct. Consequently, the NHANES results show that the symptom-versus-threshold *pattern* recurs under a related distress construct in a different population; they do not, and cannot, validate the magnitude or existence of the perceived-stress effect estimated in KNHANES. We therefore frame NHANES as supporting evidence for the pattern, and reserve genuine cross-national external validation of a fitted model for the prospective plan (H2). With that framing: The real-data demonstration is encouraging and internally coherent. The psychological-distress proxy (a positive PHQ-9 screen) was associated with both tinnitus (OR 1.61) and, more strongly, audiometric hearing loss (OR 2.29), while occupational noise was associated with tinnitus (OR 1.59) but not significantly with better-ear hearing loss (OR 1.22, $p = 0.14$), and age dominated hearing loss (OR 1.11 per year) but was unrelated to tinnitus. This dissociation—distress and noise relating to tinnitus, while age and distress relate to audiometric thresholds—mirrors the asymmetry the framework anticipated between symptom-level and threshold-level outcomes, and the sex pattern (higher tinnitus, lower hearing loss in women) is consistent with prior epidemiology. Three caveats bound these results: they are cross-sectional associations, not causal effects; the exposure is a depression *proxy* that the DAG treats as a mediator, not the perceived-stress exposure itself; and better-ear PTA understates unilateral disease, so the planned worse-ear and per-ear analyses may differ. They nonetheless show the pipeline yields sensible, reproducible estimates on genuine public data and motivate the primary KNHANES stress analysis.

7.5 Clinical implications

What should an audiologist take from a cross-sectional stress–tinnitus association? The defensible reading is modest and specific. First, in adults presenting with *non-urgent, stable*

tinnitus, perceived stress is a common co-travelling factor, which supports incorporating stress and its sequelae (sleep, mood, concentration) into tinnitus counseling and cognitive-behavioral or sound-therapy approaches (Tunkel et al., 2014)—consistent with the finding that stress tracks the tinnitus *symptom* more than the measurable *threshold*. Second, and equally important as a negative implication, the data do *not* license attributing acute, sudden, or asymmetric hearing loss to “stress”: the stress association is with the symptom, not with cochlear thresholds, and acute asymmetric presentations warrant prompt audiological and otolaryngological assessment regardless of psychosocial context. Mislabeling such a presentation as stress-related is precisely the hazard the motivating vignette illustrates.

7.6 The referral-safety layer is a separate, prospective contribution

The deterministic red-flag layer is *not* a conclusion of these data; it is a prospective safety-engineering component motivated by, but not tested in, this analysis. Because SSNHL is a time-critical emergency (Chandrasekhar et al., 2019), it is designed so a low predicted risk can never suppress urgent referral—the failure mode in the motivating vignette (resolved unilateral stress-associated tinnitus, then a delayed-care sudden loss in the other ear). Its evaluation is specified in advance (Section 5) and awaits the prospective cohort; population data identify patterns, whereas the Ajou University Hospital extension can later test treatment timing, SSNHL recovery, speech-in-noise outcomes, rehabilitation, and tinnitus burden longitudinally.

7.7 Comparison with prior work

Unlike single-country prevalence studies (Park et al., 2014; Joo et al., 2015), STARS prespecifies cross-national transportability testing with calibration, recalibration, common support, and case-mix standardization—not discrimination alone. Unlike opaque clinical-AI demonstrations, it locks feature/outcome definitions before validation, declares model intended use (Table S1), publishes model cards, and treats subgroup fairness and safety evaluation as first-class results.

7.8 Limitations

The primary hypothesis test—the KNHANES perceived-stress analysis—is now reported and supports H1, with NHANES as consistent supporting evidence; the external-validation, AI-extraction, and clinical components remain prospective, so much of the framework’s value still rests on the rigor of the prespecified plan. Several limitations bound the primary result: the KNHANES analysis is cross-sectional; the audiometry outcomes come from the ≥ 40 ENT subsample analyzed with the distributed interview-plus-examination weight (`wt_itvex`), as no ENT-specific weight is released; and the perceived-stress item is a single general question rather than a validated job-strain instrument. Substantively: (1) **measurement**—the exposure and outcome are single self-report items (no PSS-10/JCQ, no THI/TFI), so the exposure is perceived stress with work-related factors as covariates, and non-differential misclassification biases associations toward the null (detailed above); low measurement-comparability variables are excluded from cross-national pooling (Table 6); (2) **directionality**—the cross-sectional design cannot establish temporal order or exclude reverse causation (tinnitus→stress), as discussed at length above; no causal claim is made despite the DAG-guided adjustment; (3) **audiometric protocol**—the KNHANES ENT exam provides pure-tone air-conduction thresholds, and where tympanometry/bone-conduction data are unavailable to exclude conductive patterns, conductive contamination may misclassify the hearing-loss outcome; better-ear PTA further understates unilateral/asymmetric disease, motivating the reported worse-ear and high-frequency sensitivity analyses; (4) treating depressed mood/sleep as mediators rests on assumptions that may be violated, hence the separate confounder-treatment sensitivity analysis; (5) restricting to employed adults risks a healthy-worker/selection-collider effect, hence the all-adult primary analysis; (6) capping the analytic band at 69 excludes the ≥ 70 group with the highest tinnitus/hearing burden, limiting generalizability to older adults; (7) survey-design and case-mix differences complicate transportability, addressed by the weighted/unweighted, common-support, and standardization analyses; (8) SHAP importance is descriptive and can be unstable under correlated predictors; (9) finite-set

red-flag sensitivity of 1.0 does not guarantee zero deployment misses; and (10) open medical language models require local validation, privacy review, and governance before any use, and are confined here to clinician-verified research extraction.

7.9 Future directions (out of primary scope)

Beyond the three-axis core, future work may add longitudinal occupational cohorts to strengthen temporality; a neurophysiological substudy of tinnitus subtypes using open OpenNeuro/EEG datasets; audiogram-digitization to unlock legacy paper audiograms; and community extension of the harmonization schema to additional national surveys. These are deliberately kept out of the primary AJA analyses to preserve focus.

8 Conclusion

STARS reports a survey-weighted, reproducible result: in representative Korean adults aged 40–69, perceived stress is associated with the tinnitus *symptom* but not, once occupational noise and age are accounted for, with the audiometric *threshold*. This symptom-versus-threshold dissociation was prespecified (H1), is reproducible end-to-end from committed code and data mappings, and is directionally echoed by NHANES under a related distress construct. Because the design is cross-sectional, it characterizes associations—not causes—and cannot resolve whether stress precedes tinnitus or bothersome tinnitus generates distress. Around this finding, STARS also fixes, in advance, a clinically grounded companion program: cross-national external validation with recalibration, common-support, and subgroup-fairness checks; a clinician-verified, schema-constrained AI-extraction component; and a deterministic red-flag safety layer in which any prediction model is a research risk-stratification tool with declared intended use—never a diagnostic device. By releasing code, model cards, safety tests, and a synthetic-data path, the study maximizes transparency and reproducibility while avoiding unsupported causal claims. A prospective clinical cohort (the Ajou University

Hospital extension) remains necessary to study treatment timing and recovery—above all, to protect the narrow window in which sudden hearing loss is treatable.

Declarations

Author Contributions (CRediT): *Geunsik Lim*—Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing (original draft), Visualization. *Hyun Jo*—Clinical methodology (audiology/otolaryngology), Validation, Supervision of the planned clinical extension, Writing (review & editing). Both authors read and approved this draft.

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Funding: No external funding was received for this study. Any funding for the prospective clinical extension will be disclosed at that stage.

Ethics Approval: Stages 1–2 use publicly available, fully deidentified survey microdata (KNHANES, NHANES) under each provider’s data-use terms and do not constitute new human-subjects research. The Stage-3 clinical extension will obtain institutional review board approval at Ajou University Hospital before any data collection.

Reporting Guidelines: The primary observational analysis follows STROBE ([von Elm et al., 2007](#)); the prospective prediction models follow TRIPOD+AI ([Collins et al., 2024](#)); the clinical extension will follow SPIRIT. Completed STROBE and TRIPOD+AI checklists are provided as supplementary files (`paper/checklists/STROBE_checklist.md` and `paper/checklists/TRIPOD-AI_checklist.md`) and are submitted with the manuscript.

Analysis-Plan Status and Registration: This is a survey-weighted research study whose prespecified primary KNHANES perceived-stress analyses are reported here (Section 6), with NHANES as directionally consistent supporting evidence; the external-validation, AI-extraction, and clinical-extension components remain prospective (design only). The prespecified analysis plan (cycles, variables, adjustment sets, endpoints, and metrics) is

time-stamped in the versioned repository.

Data Availability: KNHANES data are available from the Korea Disease Control and Prevention Agency subject to official access procedures; NHANES public-use files are available from the U.S. CDC. The repository provides code, configuration, variable-mapping templates, and a synthetic-data generator, but does not redistribute restricted raw data.

Code Availability: All analysis code, configuration, model cards, prompt templates, and the reporting checklist are openly available at <https://github.com/leemgs/stars-audiology> under an open-source license.

AI-Use Disclosure: Generative AI tools assisted in drafting and structuring this manuscript and the code scaffold; the authors reviewed and are responsible for all content. In the proposed study, open medical language models will be used only for research feature extraction and guideline-linked explanation from deidentified text, always with clinician verification before analysis and never for autonomous clinical decisions.

Conflicts of Interest: The authors declare no competing interests.

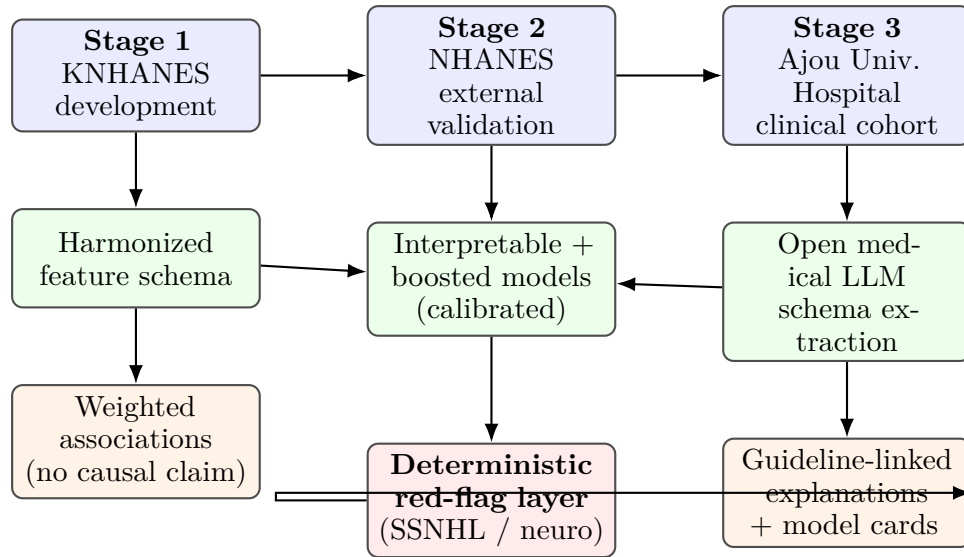


Figure 1: STARS conceptual framework. Public survey data (blue) feed a harmonized schema and calibrated, interpretable models (green); an open medical language model performs schema-constrained extraction in the clinical extension. A deterministic red-flag safety layer (red) can override probabilistic output to force urgent referral, so a low predicted risk never suppresses time-critical SSNHL care. Outputs (orange) are associations and guideline-linked explanations, not diagnoses.

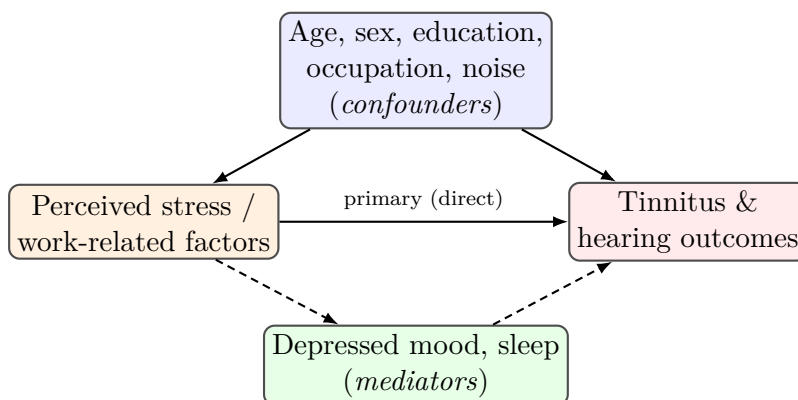


Figure 2: Assumed causal directed acyclic graph (DAG). The minimal sufficient adjustment set (blue confounders) is used in the *primary* analysis. Depressed mood and sleep (green) are treated as *mediators* of the stress→outcome path (dashed); conditioning on them (extended model) is a mediator-adjusted, potentially overadjusted estimate reported separately. Restricting to employed adults can open a selection/collider path, so all-adult analysis is primary and employed-restricted analysis is secondary.

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Table 4: Selected public datasets for STARS

Dataset	Role	Strengths	Main limitations
KNHANES	Primary development dataset	Korean population relevance; audiometry in selected cycles; tinnitus and health variables; survey weights	Cross-sectional; stress variables may be perceived stress rather than validated occupational stress; raw data access procedures required
NHANES	External validation dataset	Public CDC files; audiometry and noise exposure in selected cycles; rich health covariates; geographic validation	U.S. population; cycle-specific hearing and tinnitus variables; complex survey design
OHHR / Zenodo audiology data	Secondary method testing	Audiology-specific measurements; useful for model benchmarking	May lack stress, occupation, or tinnitus variables
OpenNeuro tinnitus datasets	Optional neurophysiology substudy	Public neuroimaging or fNIRS data for tinnitus biomarkers	Not designed for occupational stress or longitudinal treatment outcomes

Table 5: Target harmonized variables, roles, and availability. Exposure and outcome roles follow the naming principle and the DAG (Figure 2): perceived stress is the exposure; work-related factors and the minimal confounder set are covariates; depressed mood and sleep are treated as mediators in the primary analysis. Availability is cycle-dependent; see Table 6 for source items and comparability.

Harmonized variable	Role	KNHANES	NHANES
Tinnitus presence	Outcome	Sel. cycles	Sel. cycles
Bothersome tinnitus	Outcome	Sel. cycles	Sel. cycles
Better-ear PTA _{0.5,1,2,4}	Outcome	Sel. cycles	Sel. cycles
Worse-ear PTA & per-ear thresholds	Outcome	Sel. cycles	Sel. cycles
Asymmetric hearing loss	Outcome	Sel. cycles	Sel. cycles
Self-reported hearing difficulty	Outcome (2 nd)	Yes	Yes
Hearing-aid use	Outcome (2 nd)	Sel. cycles	Sel. cycles
Perceived stress	Exposure	Yes	Yes
Work-related factors (occupation, employment, hours)	Covariate / modifier	Yes	Yes
Occupational + recreational noise (duration/intensity)	Confounder	Sel. cycles	Yes
Depressed mood	Mediator	Yes	Yes
Sleep duration	Mediator	Sel. cycles	Sel. cycles
Smoking, alcohol, cardiometabolic disease	Confounder	Yes	Yes
Age, sex, education	Confounder	Yes	Yes
Socioeconomic status; race/ethnicity	Fairness axis	Partial	Yes
Survey weight, stratum, PSU	Design	Yes	Yes

Table 6: Cross-national harmonization and measurement-comparability plan (excerpt). Exact cycle-specific variable codes and response categories are pinned in the repository data dictionary. Comparability governs use: *low*-comparability variables are excluded from the common transportable model and entered only in country-specific extended models. “Sel. cycles” = present in selected cycles only.

Harmonized variable	vari-	KNHANES (source/cycle)	NHANES (source/cycle)	Original item (abridged)	Transformation rule	Compar.
Tinnitus presence		Otologic Q, 2010–2012	Audiometry Q (AUQ), sel. cycles	“Ringing/noise in ears past 12 mo?”	Any vs. none (binary)	High
Bothersome tinnitus	tinni-	Otologic Q, 2010–2012	AUQ bother item, sel. cycles	Annoyance / sleep / concentration	Bothersome vs. not	Medium
Pure-tone thresholds	thresh-	Audiometry exam, 2010–2012	Audiometry exam, sel. cycles	Air-conduction dB HL per freq.	Better/worse-ear PTA; per-ear	High
Perceived stress		Health Q (stress item)	DPQ / stress-related item	“How much stress in daily life?”	Collapse to high vs. low	Low
Depressed mood		PHQ-9 / mood item	PHQ-9 (DPQ)	Standardized depression items	Screen-positive vs. negative	Medium
Occupation / employment		Socioecon. Q	Occupation (OCQ)	Job category, employed y/n	Common category crosswalk	Medium
Occupational noise		Sel. cycles	Noise exposure (AUQ)	Loud-noise exposure, duration	Duration/intensity + binary	Medium
Sleep duration		Health Q	Sleep (SLQ)	Average hours of sleep	Continuous hours	Medium
Smoking / alcohol		Health Q	SMQ / ALQ	Current status, quantity	Standard categories	High
Age, sex, education		Demographics	Demographics (DEMO)	—	Direct / cross-walk	High
Survey weight, stratum, PSU		Design vars	Design vars	—	Kept for design-consistent SEs	High

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Supplementary Materials: Prospective AI/Safety Component

These tables specify the prespecified design of the prospective AI-extraction and deterministic referral-safety component (Section 5); they carry *no results* and are provided as supplementary material.

Table S1: Intended-use specification for the two prediction models. Both are *research risk-stratification* tools only; neither is a screening, diagnostic, or triage device. The specification is fixed before validation so that performance is judged against a declared purpose.

Dimension	Specification
Purpose	Research risk stratification for cohort description and hypothesis generation; <i>not</i> individual screening, diagnosis, triage, or treatment guidance.
Target user	Researchers/epidemiologists analyzing survey or, later, deidentified cohort data.
Prediction time	Cross-sectional (concurrent survey visit); no temporal/prognostic claim.
Target population	Adults 40–69 y in the fixed audiometry windows; not generalized beyond.
Inputs	Prespecified harmonized covariates (Table 5); no free-text or image inputs in the public-data models.
Output / decision	Calibrated probability for research stratification; drives <i>no</i> clinical action.
Cost of false positive	Low in research use; would be non-trivial if misused for screening (over-referral) — such use is contraindicated.
Cost of false negative	Low in research use; dangerous if misused clinically (missed disease) — such use is contraindicated.
Contraindicated uses	Individual diagnosis, treatment decisions, insurance/employment decisions, or any deployment without prospective clinical validation and regulatory review.
Safety interaction	The deterministic red-flag layer (Table S4) overrides model output for time-critical symptoms regardless of predicted probability.

Table S2: AI components and safety constraints

Component	Intended use	Safety constraint
Survey harmonization	Map KNHANES/NHANES variables to common schema	Human-readable mapping file and version control
Statistical baseline	Estimate adjusted associations and transparent risk scores	Survey-aware methods; no causal claims from cross-sectional data
MedGemma or similar open medical LLM	Extract fixed fields from deidentified clinical notes in future cohorts	Clinician verification; no diagnosis or treatment recommendation
Retrieval-augmented explanation	Draft evidence-linked explanations for researchers and clinicians	Curated sources only; unsupported outputs flagged
Red-flag rule layer	Identify sudden hearing-loss or neurologic warning signs	Deterministic referral rules override probabilistic model

Table S3: Prespecified evaluation metrics (frozen before external validation). The right column shows *synthetic-scaffold* values that verify pipeline wiring only: they are computed on randomly generated data, carry **no** epidemiological meaning, and must not be read as results. Real-data estimates replace them when KNHANES/NHANES files are supplied.

Dimension	Metric(s)	<i>Synthetic scaffold (not a result)</i>
Discrimination	AUROC, AUPRC (unweighted and survey-weighted)	Tinnitus: 0.64 / 0.49
Calibration	Calibration-in-the-large and calibration slope, reported separately, before and after recalibration; Brier score; plots	Tinnitus Brier: 0.20
Transportability	Common-support region; case-mix-standardized performance; reduced-common vs. country-specific model	Reported per target
Clinical utility	Decision-curve net benefit	Reported per target
Interpretability	SHAP importance (descriptive; caveated for correlated predictors)	Reported per target
Fairness	Subgroup AUROC & calibration: sex, age, employment, noise, asymmetry, SES, race/ethnicity	Reported per subgroup
LLM extraction	Exact match; precision/recall/F1; omission; span agreement; temporal/laterality/onset/duration accuracy; doc-level match; run consistency; clinically significant error rate	Clinical extension
Safety layer	Sensitivity (95% CI); specificity/over-referral; subgroup & robustness (Table S4)	Target $\rightarrow$ 1.0

Table S4: Prespecified evaluation plan for the deterministic red-flag (SSNHL/neurologic) safety layer. Sensitivity (red-flag recall) is the primary metric because a missed red flag is the costly error; but a perfect score on a finite test set does not guarantee zero real-world misses, and this limitation is stated explicitly.

Evaluation element	Prespecification
Vignette set	≥ 200 vignettes; composition fixed: real deidentified clinical cases, expert-authored cases, and synthetic cases in a stated ratio (e.g., 40/30/30%).
Reference labeling	Each vignette independently labeled by ≥ 2 otolaryngology/audiology experts; disagreements adjudicated.
Inter-rater agreement	Reported (Cohen's/Fleiss' κ) before adjudication.
Primary metric	Sensitivity (red-flag recall) with 95% CI.
Secondary metrics	Specificity and over-referral rate with 95% CIs.
Subgroup performance	By laterality (unilateral/bilateral), onset (acute/gradual), and vertigo/neurologic signs.
Robustness	Negation, ambiguous temporal expressions, and colloquial patient phrasing explicitly tested.
Error analysis	Every missed red flag audited and reported as a critical failure.
Stated limitation	Finite-set sensitivity of 1.0 is a target, not a guarantee of zero misses in deployment.